

SECTION TWO

Brainstorming and Beyond

CHAPTER 9

Better Brainstorming

by Hal Gregersen

About 20 years ago, I was leading a brainstorming session in one of my MBA classes, and it was like wading through oatmeal. We were talking about something that many organizations struggle with: how to build a culture of equality in a male-dominated environment. Though it was an issue the students cared about, they clearly felt uninspired by the ideas they were generating. After a lot of discussion, the energy level in the room was approaching nil. Glancing at the clock, I resolved to at least give us a starting point for the next session.

“Everyone,” I improvised, “let’s forget about finding answers for today and just come up with some new questions we could be asking about this problem. Let’s see how many we can write down in the time we have left.” The students dutifully started to throw out questions, and I scribbled them on a chalkboard, redirecting anybody who started to suggest an answer. To my surprise, the room was quickly energized. At the end of the session, people left talking excitedly about a few of the questions that had emerged—those that challenged basic assumptions we had been making. For instance, were there grassroots efforts we could support rather than rules that we would hand down from the top? And instead of automatically looking elsewhere for best practices, what could we learn from pockets within our own organization that had achieved equality? Suddenly, there was much more to discuss, because we had opened up unexpected pathways to potential solutions.

Brainstorming for questions, not answers, wasn’t something I’d tried before. It just occurred to me in that moment, probably because I had recently been reading sociologist Parker Palmer’s early work about creative discovery through open, honest inquiry. But this technique worked so well with the students that I began experimenting with it in consulting engagements, and eventually it evolved into a methodology that I continue to refine. By now I’ve used it with hundreds of clients, including global teams at Chanel, Danone, Disney, EY, Fidelity, Genentech, Salesforce, and dozens of other companies; nonprofit organizations; and individual leaders I’ve coached.

Underlying the approach is a broader recognition that fresh questions often beget novel—even transformative—insights. Consider this example from the field of psychology: Before 1998, virtually all well-trained psychologists focused on attacking the roots of mental disorders and deficits, on the assumption that well-being came down to the absence of those negative conditions. But then Martin Seligman became president of the American Psychological Association, and he reframed things for his colleagues. What if, he asked in a speech at the APA’s annual meeting, well-being is just as driven by the *presence* of certain *positive* conditions—keys to flourishing that could be recognized, measured, and cultivated? With that question, the positive psychology movement was born.

Brainstorming for questions rather than answers makes it easier to push past cognitive biases and venture into uncharted territory. We’ve seen this dynamic in academic studies—in social psychologist Adam Galinsky’s research on the power of reframing during times of transition, for instance. Yet lingering in a questioning mode doesn’t come naturally to most people, because we’re conditioned from an early age to just keep the answers coming.

The methodology I've developed is essentially a process for recasting problems in valuable new ways. It helps people adopt a more creative habit of thinking and, when they're looking for breakthroughs, gives them a sense of control. There's actually something they can do other than sit and wait for a bolt from the blue. Here, I'll describe how and why this approach works. You can use it anytime you (in a group or individually) are feeling stuck or trying to imagine new possibilities. And if you make it a regular practice in your organization, this type of brainstorming can foster a stronger culture of collective problem-solving and truth-seeking.

What Process Should We Follow?

Over the years, I have tested variations of this brainstorming process—I now call it a *question burst*—and collected and analyzed participant data and feedback to gauge what works best. I've experimented with different group sizes, time allotments, and numbers of questions; impromptu versus scheduled sessions; various modes of capturing ideas; and greater and lesser amounts of coaching (on, for example, what constitutes a good question and how to make creative leaps in thinking). I've done temperature checks in sessions and conducted surveys after them, looking for the effects of each variation. Over time the question burst has settled into a standard format, which consists of three steps.

1. Set the stage

To begin, select a challenge you care deeply about. Perhaps you've suffered a setback or you have a vague sense of an intriguing opportunity. How do you know it's ripe for a breakthrough question? It's probably a good candidate if it "makes your heart beat fast," as Intuit's former chair and CEO, Brad Smith, put it to me. You'll give it your full attention and want to engage others in thinking about it.

Invite a few people to help you consider that challenge from fresh angles. Though you can do this exercise on your own, bringing others into the process will give you access to a wider knowledge base and help you maintain a constructive mindset. As Ned Hallowell says in his book *Driven to Distraction at Work* (which was based on his decades of research on how to sustain productive attention), worry "feasts on a solitary victim." When you ask others to participate in a question burst, you're making yourself vulnerable by sharing the problem—but you're also summoning empathy, which fosters idea generation, as we've learned from design thinking. And you engage others in the cause in a nonthreatening way.

It's best to include two or three people who have no direct experience with the problem and whose cognitive style or worldview is starkly different from yours. They will come up with surprising, compelling questions that you would not, because they have no practiced ways of thinking about the problem and no investment in the status quo. They're more likely to ask third-rail questions and point to elephants in the room—they don't know not to.

In traditional brainstorming—the kind that focuses on generating answers—individuals perform better than groups, on average. That's because powerful group dynamics such as social loafing (coasting on others' contributions) and social anxiety (fears about how your ideas will be judged) can hinder original thinking and stifle the voices of introverted members. But the question-burst methodology, by design, reverses many of those destructive dynamics by prompting people to depart from their usual habits of social interaction. For one thing, it creates a safe space for anyone, including a quieter person, to offer a different perspective. Because a question burst doesn't demand that anyone instantly assert a point of view, people often feel more comfortable speaking up. The sole focus on questions also suspends the automatic rush to provide an answer—and ultimately helps expand the problem space for deeper exploration.

Once you've gathered your partners for this exercise, give yourself just two minutes to lay out the problem for them. People often believe that their problems require detailed explanations. But quickly sharing the challenge forces you to frame it in a high-level way that doesn't constrain or direct the questioning. So just hit the highlights. Try to convey how things would change for the better if the problem were solved. And briefly say why you are stuck—why it hasn't already been solved.

This approach helped Odessa, a manager at a global financial services company, reframe what she initially viewed as a complex communications challenge: rolling out a new strategy to people performing different tasks at many levels across many geographic locations. She prefaced her question burst with a simple explanation, sharing her hopes for getting everyone “rowing in the same direction” and her frustration that one set of messages couldn't do the job, given employees' diverse roles and perspectives. With this simple explanation, she created room for a line of questioning that radically altered her understanding. She came to see the problem as a leadership challenge, not just an internal marketing campaign. If she could find a way to trust others to convey the strategy, she could mobilize a small army of managers in the field to tailor messages for maximum local impact.

NOT ALL QUESTIONS ARE CREATED EQUAL

Often, as I'm outlining the rules for a question burst, people ask what kinds of questions they should contribute—or how they can be confident that a question is a good one for further pursuit. While I hesitate to be definitive about this, not all questions have equal potential to lead to novel solutions. To up your odds, keep these principles in mind:

- Traditional divergent-thinking techniques (for example, making random associations or taking on an alternative persona) can help unlock new questions and, ultimately, new territory.
- Questions are most productive when they are open versus closed, short versus long, and simple versus complex.
- Descriptive questions (What's working? What's not? Why?) work best if they precede speculative ones (What if? What might be? Why not?).
- Complex questions that demand creative synthesis produce better breakthrough thinking than simple questions that require only recall.
- Questions are annoying and distracting when they don't spring from a deeply held conviction about what the group wants to achieve.
- Questions are toxic when they are posed aggressively, put people on the spot, cast unwarranted doubt on their ideas, or cultivate a culture of fear.

Before opening the floor to your group, clearly spell out two critical rules: First, people can contribute only questions. Those who try to suggest solutions—or respond to others' questions—will be redirected by you, the convener of the session. And second, no preambles or justifications that frame a question will be allowed, because they'll guide listeners to see the problem in a certain way—the very thing you're trying to avoid.

You'll also want to do a quick emotion check up front. As the owner of the challenge, take a moment to reflect on it: Are your feelings about it positive, neutral, or negative? Jot down a few words that capture your own baseline mood. No need to spend more than 10 seconds on this. You'll do the same thing again after the session is over. These checks are important because emotions affect creative energy. The exercise's objective is not only to spark valuable new questions but also to provide an emotional boost that will make you more likely to follow up on them.

Here I should point out that your creative energy will ebb and flow in the coming days, weeks, and months—and preparing yourself for that is critical. Transformational ideas start out as exhilarating but turn vexing as unforeseen snags reveal themselves. Then they settle into hard work that, with luck, produces moments of hope that will see the change through. If you expect this turbulence from the beginning, you'll be better able to ride it out later.

2. Brainstorm the questions

Now set a timer and spend the next four minutes collectively generating as many questions as possible about the challenge. As in all brainstorming, don't allow pushback on anyone's contributions. The more surprising and provocative the questions, the better.

When working with large enterprises, I often notice that senior leaders in particular find it excruciatingly difficult to resist offering answers—even for four minutes—when people start throwing out questions. At one manufacturing company, for instance, when questions about supply chain issues started bubbling up, the group's leader couldn't help jumping in to display his knowledge. This impulse is understandable, and not just for senior executives. In a hierarchy, any manager's failure to have ready answers may be perceived as an embarrassing stumble. Questions, especially counterintuitive ones, make many of us feel so uncomfortable that we hasten to utter any default response that buys us time to recover. But when we're feeling blocked on a problem, answering questions this way is a waste of time. After all, the reason we're hung up is that our go-to answers aren't getting us anywhere.

In this exercise, the emphasis is on quantity. By asking the group to generate as many questions as possible in the time allotted—try for at least 15 questions—you'll keep them short, simple, and fresh. Write every question down verbatim on paper or type it into a laptop or a tablet instead of using a whiteboard. This way, you can capture everything accurately. And ask group members to keep you honest afterward. Otherwise you might commit unconscious censoring that repels lines of inquiry you don't immediately understand or want to hear.

As you're recording, add your own questions to the mix. Doing so will often reveal patterns in how you have habitually framed a problem (and might have unknowingly perpetuated it).

Is there some magic about precisely four minutes and 15 questions? No, but the time pressure helps participants stick to the questions-only rule. Any effort spent on answers will mean less chance of hitting the goal. People will also be more likely to generate questions that are unburdened by qualifications and assumptions, and they'll find it easier to resist explaining why they're asking a question that might seem to come from left field. Even better, studies show that moderate performance pressures can enhance creative output.

Moreover, perhaps because selective, sustained attention places real demands on the human brain, energy often wanes in this exercise after three and a half minutes, especially for beginners. And as a practical matter, transcribing dozens of questions can turn into an onerous task. For both those reasons, it's better to use multiple question bursts to reshape, refine, and ultimately solve a challenge than to cram too much activity into one longer session.

Once the timer goes off, do a second quick emotional check. How do you feel about the challenge now? (And how do others in the group feel about it?) Are you more positive than you were four minutes ago? If not, and if the setting allows, maybe rerun the exercise. Or get some rest and try again tomorrow. Or try it with some different

people. Research has established that creative problem solving flourishes when people work in positive emotional states. After poring over survey data from more than 1,500 global leaders, I'm convinced that part of the power of the question burst lies in its ability to alter a person's view of the challenge, by dislodging—for most—that feeling of being stuck.

3. Identify a quest—and commit to it

On your own, study the questions you jotted down, looking for those that suggest new pathways. About 80% of the time, this exercise produces at least one question that usefully reframes the problem and provides a new angle for solving it. Select a few questions that intrigue you, that strike you as different from how you've been going about things, or even that cause you to feel a bit uncomfortable.

Now try expanding those few into their own sets of related or follow-up questions. A classic way of doing this is the five-whys sequence developed by Toyota Industries' founder, Sakichi Toyoda—or its variation suggested by Stanford University's Michael Ray in his book *The Highest Goal*. Ask yourself why the question you chose seemed important or meaningful. Then ask why the reason you just gave is important—or why it's a sticking point. And so on. By better understanding why a question really matters and what obstacles you might face in addressing it, you deepen your resolve and ability to do something about it and further broaden the territory of possible solutions. In the case of Odessa, the manager with a strategy to roll out, one breakthrough question—Could you recruit field leaders to communicate it locally?—provoked other questions: Why haven't I done that in the past? Could I trust others to do this well? Why do I have a problem extending that trust?

Finally, commit to pursuing at least one new pathway you've glimpsed—and do so as a *truth seeker*. I steal that term from *The Right Kind of Crazy*, NASA engineer Adam Steltzner's account of working at the Jet Propulsion Laboratory, where freewheeling people manage to accomplish things like landing a robotic rover on Mars. Set aside considerations of what might be more comfortable to conclude or easier to implement, and instead adopt an innovator's focus on the "job to be done" and what it will take to get the problem solved. Devise a near-term action plan: What concrete actions will you personally take in the next three weeks to find potential solutions suggested by your new questions?

After one question burst I helped facilitate, a chief marketing officer from a multidivisional company resolved to track down some facts. He had been wrestling with concerns about hypercompetitive behaviors in his business unit. In a question-burst session he led with others, it dawned on him that he had been making a big assumption: that the founders of his division had chosen its unique compensation scheme to create a culture of internal rivalry. His to-do list started with getting on their calendars and asking them about this. Guess what? Not only was this not a culture they had aimed for, but they were dismayed to learn it existed. His meetings with them gave rise to a new emphasis on culture and values in the unit—and created the context in which the chief marketing officer could intervene and address toxic behaviors. The point here is that arriving at assumption-challenging questions is essential but never sufficient. An action plan and follow-up can clarify the problem and break open the pathway to change.

How Can We Make Question Bursts a Habit?

I usually recommend doing at least three rounds of the question-burst exercise for a given issue. Although it's valuable as a one-off activity, the more you do it, the deeper you'll go in your thinking. After the leader of a development team at a global software company did the exercise repeatedly, she came to the realization that her original conception of a problem was, in her words, "superficial." Through persistent questioning, she told me, she "arrived at a much more meaningful challenge to conquer."

Even with three rounds, the time investment is minimal. It's an efficient path to fresh perspectives and creativity. The process will also get easier the more you do it. When people crank up their questioning activity for the first time with this approach, it feels strange because it's out of line with established norms at work and in life. Since childhood, they've been conditioned not to ask questions.

James T. Dillon, an education professor emeritus at the University of California, Riverside, spent a career studying this phenomenon in classrooms. He was shocked by how rarely students asked questions—which are critical to learning. The problem wasn't a lack of curiosity. "Every time that conditions have been provided for them (not by a mere pause, 'Any questions? No? OK, open your books'), a flood of intriguing student questions has poured forth," Dillon writes in his book *Questioning and Teaching*. When he surveyed other teachers about this, they almost universally agreed that "students indeed have questions but do not go on to ask them in class." Why not? They're afraid to do so, Dillon says, "largely because of their experience with negative reactions from the teacher (and from classmates)." They learn to keep their questions to themselves and to repeat back well-rehearsed answers when quizzed by their teachers, according to Tony Wagner, a senior fellow at the Learning Policy Institute. Other researchers—looking at arenas of human learning and interaction such as community forums, medical consultations, political institutions, and workplaces—have consistently come to the same conclusion: Questioning is an innate human behavior that's actively subverted and systematically shut down.

And power struggles don't help matters. In social groups, dominant individuals inevitably emerge; left unchecked, they find ways to build and perpetuate their power. One common way to do this is to silence questioners—those pesky, curious minds whose queries might suggest that the leader hasn't quite figured it all out.

Of course, many business leaders, recognizing the imperative for constant innovation, do try to encourage questions. But their employees have already internalized the habit of not asking them—especially the tough ones. We need to change this habit. That's what my MIT colleague Robert Langer, the health-care technology innovator who has been called the "Edison of medicine," has been doing with his students and postdocs. In an interview he said, "When you're a student, you're judged by how well you answer questions. Somebody else asks the questions, and if you give good answers, you'll get a good grade. But in life, you're judged by how good your questions are." As he mentors people, he explicitly focuses their attention on making this all-important transition, knowing that "they'll become great professors, great entrepreneurs—great something—if they ask good questions."

Organizations can raise their questioning quotient in various ways. For example, in my field experience, I've found that people become better questioners in environments where they're encouraged to value creative friction in everyday work. At companies like Amazon, ASOS, IDEO, Patagonia, Pixar, Tesla, and Zappos, for example, people often come together to tackle challenges by asking one another tough questions—in hallways, lunchrooms, or even conference rooms. Research by management professors Andrew Hargadon of University of California, Davis, and

Beth Bechky of New York University shows that those volunteering ideas in such companies do not mindlessly spit back answers to the questions posed. People respectfully build on the comments and actions of others, considering “not only the original question but also whether there is a better question to be asked.” As they do this over and over, new solutions emerge.

People also become better questioners in organizational cultures where they feel safe doggedly pursuing the truth, no matter where it takes them. To create such cultures, MIT’s Ed Schein says, leaders must show humility, vulnerability, and trust, and they must empower others and treat them equitably. When those conditions aren’t present, questions tend to be constrained or, worse, crushed.

Interestingly, when I’ve facilitated question bursts with large groups (broken down into subgroups of three to six people), I have noticed that the people least likely to engage in the exercise and follow the rules are the folks with the highest positions or greatest technical expertise. Whether they feel they’re above the exercise or worry that sharing problems will make them appear incompetent, they cripple the truth-seeking capability of the entire group as others watch them disengage or scoff at contributions. If that’s the example and tone that leaders are setting in a single microcosmic exercise, imagine the dampening impact they have on inquiry throughout their organizations.

Finally, people must hold themselves accountable for follow-up. Few things are more annoying than a colleague who *only* asks questions. People must take responsibility for exploring the pathways those questions open up and discovering valuable answers. This is especially true for leaders. Everyone else is taking cues from them about when, where, how, and why the status quo should be challenged. Leaders must carve out time to help gather and analyze newer, better, and different information. It’s a sign of ownership when leaders go out of their way to do that. It shows others that management is committed to crafting a future where questions count.

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CHAPTER 10

The Problem-Solving Process That Prevents Groupthink

by Art Markman

There are two reasons most of us aren't very good at creative problem-solving. First, few people get training in how to be creative. Second, few people understand group dynamics well enough to harness their own power to help groups maximize their creativity.

Resolving the first issue requires getting your employees to learn more about the way they think . . . a tall order for managers. The second issue, though, is well within your ability to change.

A key element of creativity is bringing existing knowledge to bear on a new problem or goal. The more people who can engage with that problem or goal, the more knowledge that is available to work on it. Unfortunately, much research demonstrates that the traditional brainstorming methods—first described by Alex Osborn in the 1950s—fail. When groups get together and simply start throwing out ideas, they actually come up with fewer ideas overall and fewer novel, actionable ideas than what the same individuals would have generated had they worked alone.

Managing Divergent and Convergent Thinking

To fix this problem, we need to think about the two phases of group problem-solving: *divergence* and *convergence*.

Divergence happens when the group considers as many different potential solutions as possible. For example, a common test of creativity is the alternative-uses exercise. People are to try to create as many unique uses as they can.

Convergence happens when multiple proposed solutions are evaluated. In this phase, a large set of ideas is whittled to a smaller set of candidate solutions to the current problem.

The core principle of group creativity is that individuals working alone diverge, whereas group members working together converge. In group settings, as soon as one person shares a potential solution with everyone else, this idea influences the memory of every person in the group in ways that make everyone think about the problem more similarly. That is why groups working together diverge less than do individuals working alone.

To generate ideas, diverge first.

To fix group idea generation, then, be aware of when you are trying to diverge and when you are trying to converge. For example, early in the process of problem-solving, think carefully about the problem itself. Have your group members work alone to craft statements describing the problem. Then, get them back together to discuss their descriptions. The individuals are likely to come up with a variety of distinct problem statements. The group discussion will lead everyone to accept one variant or a few variants of these statements to work on. This individual, then group approach creates healthy convergence.

To generate solutions, *also* diverge first.

When you start to generate solutions, you again want to begin with divergence. As you did for problem descriptions, have people work alone to start. Then collect these initial ideas, send them around to other group members, and allow the divergence to continue as group members individually build on the ideas of their colleagues. Because people are still working alone, the way they build on other people's ideas is still going to be different from how other group members are building on those ideas.

After this process, you can give the resulting ideas to everyone and then let the group get together to discuss them. This discussion will gradually lead the group to converge on a small number of candidate solutions.

This process maximizes the contribution of the group. Everyone gets to engage their knowledge in service of the problem to be solved without having their memories or opinions influenced by other people's solutions. Everyone is also able to enhance the ideas generated by their colleagues. Finally, the group gets to work together to build further on the ideas and to evaluate the candidates.

This simple procedure works effectively, because it respects what individuals and groups do best.

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CHAPTER 11

A Smarter Way to Wrap Up Your Brainstorm

by Angus Fletcher

Almost every business, of every size, across sectors, employs creativity training, from whiteboard brainstorming sessions to design thinking. It's a billion-dollar industry. But there's a problem: The training doesn't work. Instead, it perpetuates expert bias and pseudo-innovation. And although it can temporarily boost morale, it does little over the long haul to reduce burnout. On the whole, research has shown creativity training to be at best inadequate and at worst counterproductive.¹

To understand what's broken and how to fix it, my lab partnered with teams at a variety of organizations, among them Silicon Valley startups, U.S. Special Operations, the University of Chicago Booth School of Business, and *Fortune* 50 companies. One of our key findings had to do with how you *finish* a brainstorm.

Most brainstorming sessions wrap up with the group's attempting to select the best ideas on the whiteboard. When you do that, you're actually attempting to eliminate the worst ideas through logical techniques such as convergent and critical thinking.

This wrap-up approach is counterproductive. It reinstates whatever biases you managed to escape during the brainstorm, and it kills your most promising creations. Because those creations, like newborns, will be less developed than the old standbys, they will often get crossed off the whiteboard as imperfect or impractical. What these nascent intuitions need instead is further development, via counterfactual thinking.

A more effective approach is to use this two-step, meet-the-moment process called *storythinking*.

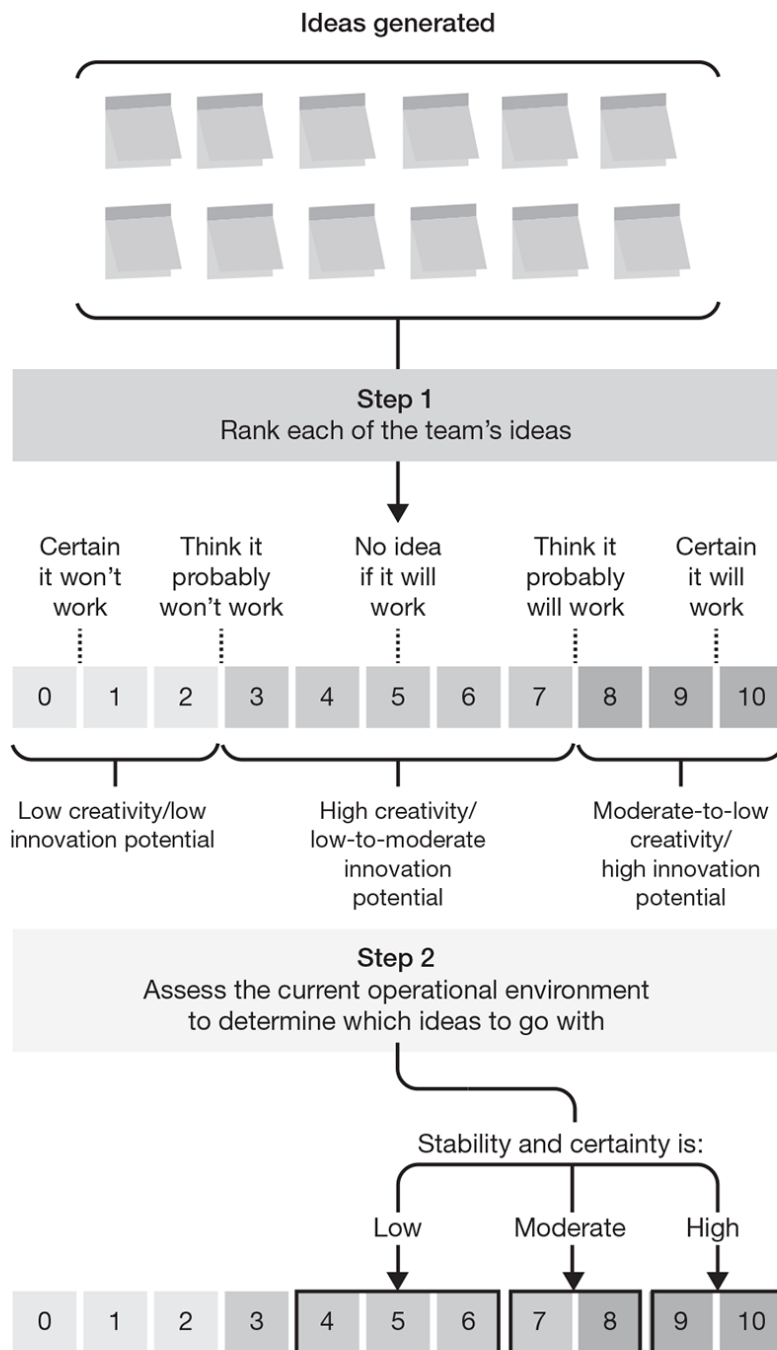
For step 1, take each of your team's newly imagined options and rank it on the scale you see in [figure 11-1](#). Options with a low ranking (0–2) are low in creativity and low in innovation potential. Options with a high ranking (8–10) are moderate-to-low in creativity but high in innovation potential. Options in the middle (3–7) are low-to-moderate in innovation potential but high in creativity.

For step 2, assess your current operational environment. Is it stable or volatile? Certain or uncertain? If it's high in stability and certainty, go with an option ranked 9 or 10. If it's moderate, go with a 7 or 8. If it's low, go with a 4 or 6. (That's not a typo. Go with a 4, an option you think might not work.) If it has no stability or certainty, go with a 5.

FIGURE 11-1

Creativity in the moment

As an alternative to brainstorming sessions for idea generation, use this two-step storythinking process to help identify and develop ideas.



In stable and certain environments, highly creative options are less likely to work, so there's no need to try them. In unstable and uncertain environments, less creative options are doomed, so the value proposition lies in gambling on a long shot. This method matches your originality to the moment.

Angus Fletcher is Professor of Story Science at Ohio State’s Project Narrative. More of his research can be found at www.angusfletcher.co.

NOTE

1. Patrick J. Kiger, “Practice Does Not Necessarily Make Perfect When It Comes to Creativity,” *Insights by Stanford Business*, September 10, 2020, www.gsb.stanford.edu/insights/practice-does-not-necessarily-make-perfect-when-it-comes-creativity.

Adapted from “3 Exercises to Boost Your Team’s Creativity” on hbr.org, March 24, 2022 (product #H06XE3).

CHAPTER 12

Find Innovation Where You Least Expect It

by Tony McCaffrey and Jim Pearson

On the evening of April 14, 1912, the RMS *Titanic* collided with an iceberg in the North Atlantic and sank two hours and 40 minutes later. Of its 2,200 passengers and crew, only 705 survived, plucked out of 16 lifeboats by the RMS *Carpathia*. Imagine how many more might have lived if crew members had thought of the iceberg as not just the cause of the disaster but a lifesaving solution. The iceberg rose high above the water and stretched some 400 feet in length. The lifeboats might have ferried people there to look for a flat spot. The *Titanic* itself was navigable for a while and might have been able to pull close enough to the iceberg for people to scramble onto it. Such a rescue operation was not without precedent: Some 60 years before, 127 of 176 passengers emigrating from Ireland to Canada saved themselves in the Gulf of St. Lawrence by climbing aboard an ice floe.

It's impossible to know if this rescue attempt would have worked. At the least it's an intriguing idea—yet surprisingly difficult to envision. If you were to ask a group of executives, even creative product managers and marketers, to come up with innovative scenarios in which all the *Titanic*'s passengers could have been saved, they would very likely have the same blind spot that the crew had. The reason is a common psychological bias—called functional fixedness—that limits a person to seeing an object only in the way in which it is traditionally used. In a nautical context, an iceberg is a hazard to be avoided; it's very hard to see it any other way.

When it comes to innovation, businesses are constantly hampered by functional fixedness and other cognitive biases that cause people to overlook elegant solutions hidden in plain sight. We have spent years investigating how innovative designs can be built by harnessing the power of the commonly overlooked. We have identified techniques and tools to help you overcome cognitive traps and solve problems in innovative ways—whether you are conceiving new products, finding novel applications for existing products, or anticipating competitive threats. Using the tools doesn't require special talents or heroic degrees of creativity; taken together, they form a simple, low-cost, systematic way to spur innovation.

To understand how the tools work, let's first look at the three cognitive barriers they address.

Functional Fixedness

In the 1930s, German psychologist Karl Duncker demonstrated the phenomenon of functional fixedness with a famous brainteaser. He gave subjects a candle, a box of thumbtacks, and a book of matches and asked them to find a way to affix the candle to the wall so that when it was lit, wax would not drip onto the floor. Many people had a hard time realizing that the answer was to empty the box of tacks, attach the candle to the inside of the box with melted wax, and then tack the box to the wall. The box acts as a shelf that supports the candle and catches the dripping wax. Because the box had been presented to subjects as a tack holder, they couldn't see it any other way.

In similar puzzles—known by cognitive psychologists as *insight problems*—people have trouble seeing that in a pinch, a plastic lawn chair could be used as a paddle (turn it over, grab two legs, and start rowing); that a basketball could be deflated, formed into the shape of a bowl, and used to safely carry hot coals from one campsite to another; or that a candlewick could be used to tie things together (scrape the wax away to free the string).

What causes functional fixedness? When we see a common object, we automatically screen out awareness of features that are unimportant for its use. This screening is an efficient neurological tactic for everyday life, but it's the enemy of innovation.

One way to overcome the problem is to change how you describe an object. When told that a candlewick is a string, for instance, almost everyone recognizes that it could be used to tie things together. Our *generic parts technique* is a systematic way to change how an object is described. By avoiding unintentionally narrowing people's conception of the object, we open them up to more ideas for its uses.

We consider each element of an object in turn and ask two questions: Can it be broken down further? And does our description imply a particular use? If the answer to either question is yes, we keep breaking down the elements until they're described in their most general terms, mapping the results on a simple tree. When an iceberg is described generically as a floating surface 200 to 400 feet long, its potential as a lifesaving platform soon emerges.

Calling something a wick implies its use as a conduit of a flame. Describing it as a string strips away a layer of preconceived uses and suggests less common ones. Breaking the string down further into its constituent parts of fibrous strands might spark even more uses.

To see if the use of generic descriptions bolsters creative thinking, our research team presented two groups of students with eight insight problems that required overcoming the functional fixedness bias for their solution. We told the members of one group simply to try their best. We taught the other group the generic-parts technique and then asked them to use it on the problems. The people in the first group solved, on average, 49% of the problems (just shy of four of them). Those who systematically created generic descriptions of their resources solved, on average, 83% (or 6.64) of them.

Design Fixation

Simple insight problems given in a psychology lab can be solved by focusing on only four types of features—materials, size, shape, and parts. But solutions to real-world engineering problems often depend on noticing unusual aspects of a broader range of features. This wide-open, imaginative outlook, as we noted, is difficult to develop.

We studied this phenomenon by asking 15 people to list as many features and associations as they could for a candle, a broom, and a dozen other common objects. We then classified their responses by the type of feature, including its color, shape, material, designed use, aesthetic properties, along with the emotions it evokes, the type of energy it generates, and the other objects that the original object is commonly paired with. On average, participants overlooked almost 21 of the 32 types of features (about 65%) that we had previously identified for each object.

Why? When handed a product and asked to create a new design or variation on it, people tend to fixate on the features of the current design. This obstacle to novelty is called *design fixation*. To take a real-world example, when people are shown a sturdy, resealable pouch full of candy and asked to think of a new design that could lead to new uses, they tend to manipulate the types of features used to create the current design. That is, they focus on the width of the base of the pouch or the rigidity of the plastic that makes it stand. To be truly innovative, however, you need to manipulate the features that everyone else has overlooked (see [figure 12-1](#)).

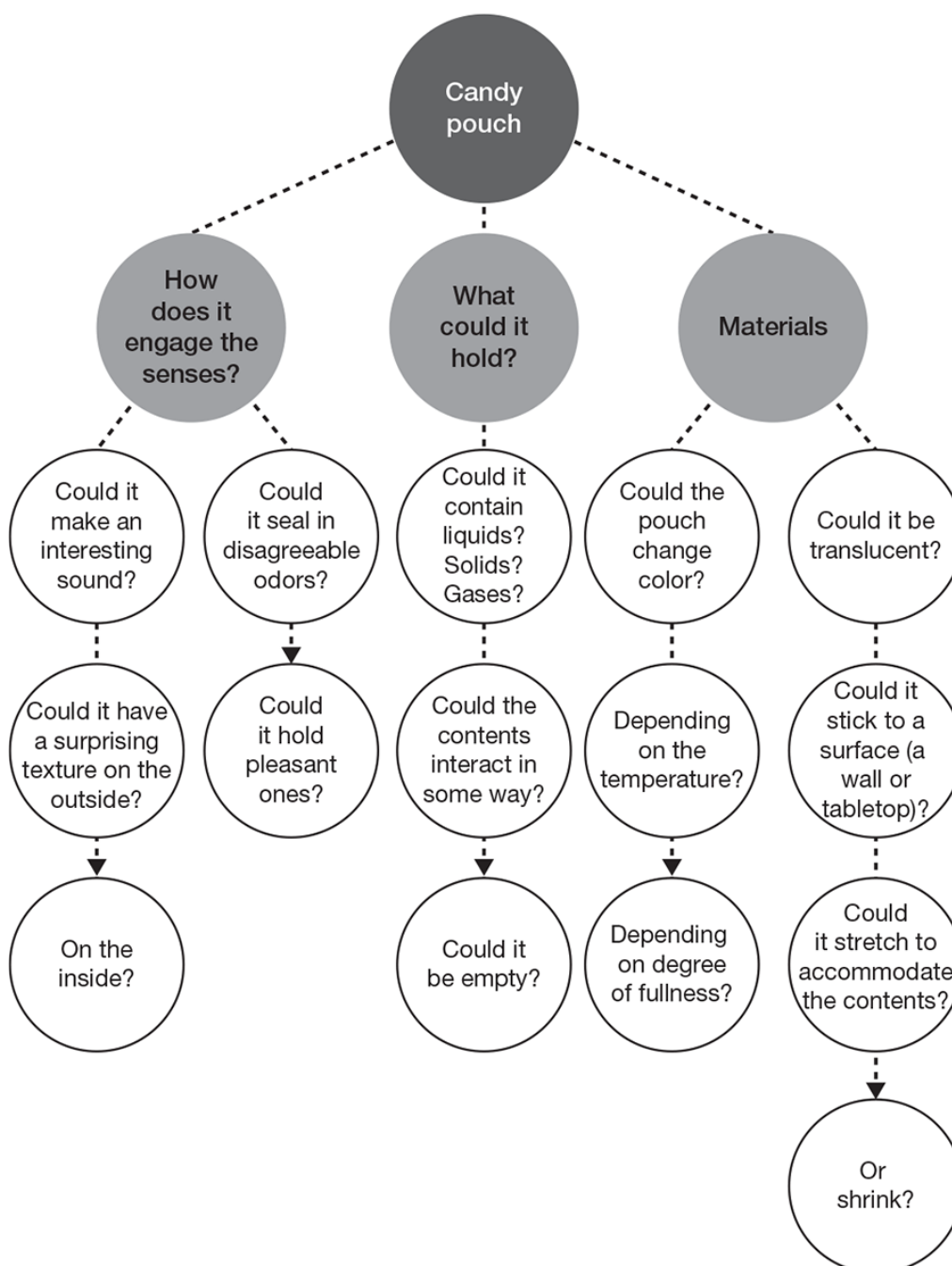
But how do you do that? Just as airline pilots have long used checklists to make sure they don't skip any necessary steps when preparing for flight, we developed a checklist of types of product features that people tend to overlook. Whether your product is a physical object or an intangible process, we recommend that you develop a checklist of features that were important to your previous and current innovation projects and add to the list with each new project. Teams working on innovation projects can then refer to the list to prompt them to consider features they would probably otherwise overlook—thus saving time, effort, and frustration.

Examining the pouch of candy with our checklist in mind permitted us to easily uncover many features that could lead to new designs and new uses. First, every pouch sold contains something. This feature is so obvious that its absence is commonly overlooked. Why not sell empty pouches so that customers can decide what to use them for: jewelry, spare change, nuts and bolts, cosmetics, and so on? Imagine empty pouches next to the sandwich bags, freezer bags, and storage bags in your supermarket. Second, most pouches sold are about the size of your hand. Systematically considering changes to the size triggers new ideas for possible contents. What about selling a gallon of paint in a resealable pouch, for instance? Third, current pouches have one inner compartment. But what might you do with more? You could, say, sell two-compartment pouches for things you want to mix together later: cereal in the top compartment and milk in the bottom, salad in the bottom and dressing in the top, and so on. Fourth, consider the pouch as a container of a scent (either a good or a bad smell). You could sell a large pouch as a garbage can that reseals to keep in the odor. These are just a few of the new designs that emerge from contemplating a checklist of overlooked features.

FIGURE 12-1

Promising features for a plastic resealable candy pouch

If you consider an object's less obvious characteristics, new purposes may arise. Here are some features to consider for a candy pouch.



Goal Fixedness

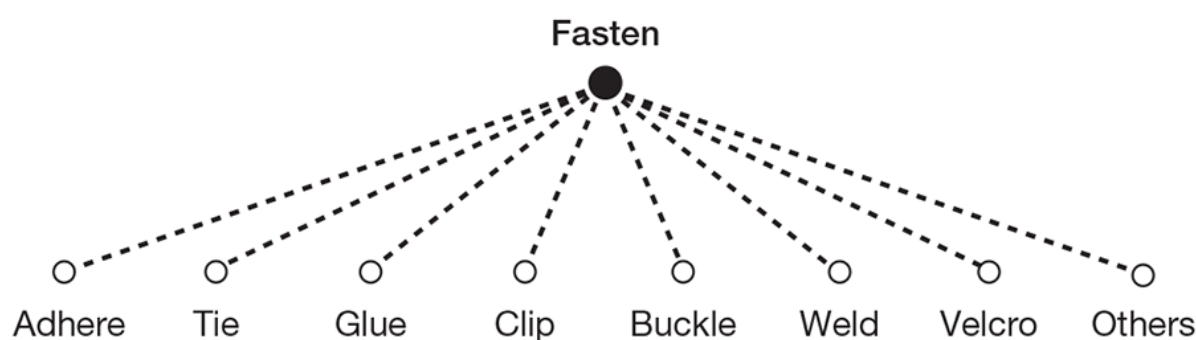
Suppose we asked you to think of a way to adhere something to a garbage can. Chances are that like most people, you would think of using glue or tape, both forms of adhesives. But what if we asked you instead to *fasten* something to the can? Just switching a specific verb like *adhere* to a more general one would most likely prompt you to list a wider range of possibilities: binder clip, paper clip, nail, string, Velcro, and so on. That's because the way a goal is phrased often narrows people's thinking. We call this barrier *goal fixedness*. Framing a problem in more general terms can help you overcome this form of fixedness.

But what constitutes a more general term? Is *fasten* more general than *adhere*? A good resource for mapping terms is a thesaurus that makes hierarchical structure explicit by identifying *hyponyms*—more-specific synonyms—for a term. For example, the online thesaurus WordNet indicates that there are at least 61 ways to fasten things—including sew, clamp, chain, garter, strap, hook, staple, belt, screw, wire, buckle, cement, tack, joggle, button, and rivet (figure 12-2). Each method describes the concept of fastening one thing to another in a slightly different way and gives rise to diverse solutions. *Adhere*, by contrast, has only four hyponyms.

FIGURE 12-2

What's in a name?

How broadly—or narrowly—you phrase a goal affects how you visualize it.



Action words, the centerpiece of most goals, often have hyponyms. Each hyponym hints at a more specific way to achieve the goal. There are 172 hyponyms for the verb *remove*, 50 for *guide*, 46 for *transport*, 115 for *separate*, and—perhaps surprisingly—only 24 for the seemingly very general term *mix*.

Of course, a goal consists of more than just a verb. The verb expresses what sort of change you're after, but nouns express what needs changing, and prepositional phrases express important constraints and relationships between things. Put them all together, and almost any goal can be expressed as a verb (*fasten*), a noun (*something*), and a prepositional phrase (*to a garbage can*). Try it: Increase sales in Massachusetts, reduce vibrations in skis, and so on. By putting your goal in this format and playing with the hyponyms of each of its parts, you can explore diverse approaches to your problem in a simple and cost-effective way.

Here's how the approach worked when one of us (Jim) applied it to the real-world goal of reducing concussions in American football. First he dropped the prepositional phrase *in football* from consideration and focused on the verb and noun: *reduce concussions*. To break free of hidden assumptions, he used WordNet to rephrase the goal in as many ways as possible: *lessen trauma*, *weaken crash*, *soften jolt*, *reduce energy*, *absorb energy*, *minimize force*, *exchange forces*, *substitute energy*, *oppose energy*, *repel energy*, *lessen momentum*, and so on. Using Google, he performed searches such as “concussions lessen trauma” to see which ways of phrasing the goal had been heavily explored already and which ones were underexplored.

Jim found that in the context of concussions, the phrase *repel energy* had relatively few search results—a sign that the solution it implied might have been overlooked. One way to repel energy is through magnets, and this idea suggested a possible approach: Make each helmet magnetic with the same pole so that two helmets would repel each other when near one another. Results from initial tests showed that when the helmets were about to collide they decelerated, and because of their circular shape, they tended to glance off each other, as two magnetic billiard balls would, rather than smashing head-on. Several physicists have verified the plausibility of this approach for significantly reducing the force during helmet collisions.

We began the patenting process for our solution, but our lawyer discovered that someone had submitted the same idea just weeks earlier. We tip our hat to that person.

Visualizing Innovative Thinking

At its most basic level, problem-solving consists of two connected activities: framing a goal and combining resources to accomplish it. Each variation of the goal and every discovery of a “hidden” feature of an available resource can suggest a different course to take. Our approach involves mapping the relationships between all the possibilities in a simple graph, somewhat analogous to a decision tree.

Starting with the goal at the top, we represent each refinement of the goal as a vector pointing downward. The available resources are placed at the bottom, with their features extending upward. Interactions between the resources and their features extend further toward the top. When the two sets of vectors connect, we have a *solution path*. Such a path can be built by working from the top down, by going from the bottom up, or by switching back and forth between the goal and the resources.

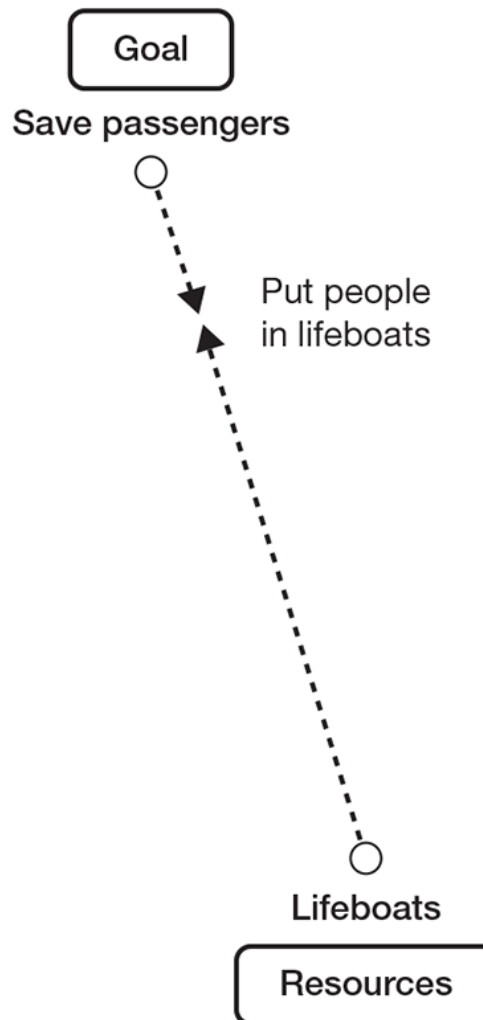
This approach is an effective alternative to traditional brainstorming sessions for group innovation work, because it allows people to play to their strengths: Strategically oriented people can focus on refining the goal, while those more familiar with technologies and production processes can begin with the resources. We call this approach *brain-swarming*—a nod to the concept of swarm intelligence. As people contribute to the growing graph, their activity resembles a swarm of insects.

To understand how this visualization approach works, let's return to the problem facing the passengers on the *Titanic*. We'll start with the goal “save passengers.” The most obvious resources are the lifeboats. The simplest application of the resources to achieve the goal is “put people in the lifeboats.” Thus, we begin with a straight line (see [figure 12-3](#)).

FIGURE 12-3

Dominant survival strategy on the *Titanic*

The first step in discovering how resources could be used to reach a goal is to map the most obvious solution.



Next, we find different ways to phrase the goal to bring out different solutions. For instance, slightly different goals would be to keep people warm and breathing and to keep people out of the water. Let's look closer at one of these options: keep people out of the water. One way to keep them out of the water is to place them on floating things—not just lifeboats. This idea might spark a fuller consideration of the resources at hand. You might realize that since wood can float, for instance, wooden tables might have been of help. Planks, or perhaps doors, from the ship might have been placed between the lifeboats to hold more people out of the water.

Moving from floating things to even more general considerations of buoyancy might bring to mind the many steamer trunks on board. Tying a set of trunks together to produce another sort of makeshift floating platform might have been enough to support several people directly or to provide a foundation on which to build a more secure platform of wooden planks.

It was estimated that as many as 40 cars were on board. That means 160 tires and inner tubes (not to mention spare tires) were at the passengers' disposal. Tying together rubber tires and inner tubes might conceivably have created a floating raft on which wooden boards could have been placed. And of course, the iceberg itself is a giant floating thing.

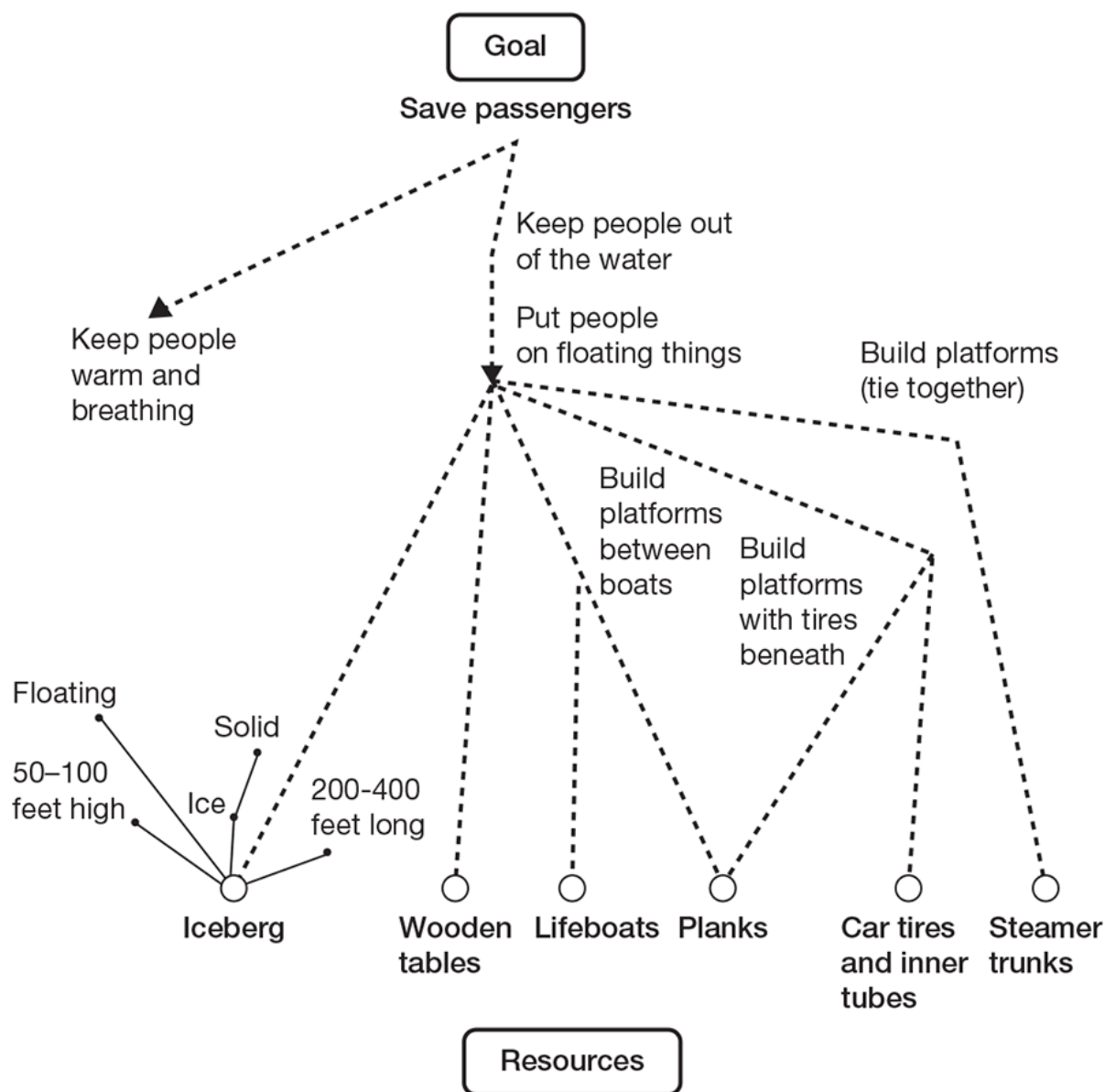
On that April night in 1912, none of these ideas might have worked, particularly since it took so long for people to understand the peril they were in. But the point of such an exercise is not to discover the right solution; it is to uncover as many connections between the goal and the widest possible view of the features of the available resources so that you can look beyond the obvious (see [figure 12-4](#)).

The goal of the brain-swarming graph, therefore, is to distill the problem-solving process to its most basic components and show how they are all related to one another. People do not have to remember all the components under consideration, because the graph shows them in a glance. This systematic approach takes some of the mystery out of innovation.

FIGURE 12-4

Overlooked strategies for saving *Titanic* passengers

Find new ways to name the goal, and new resources may present themselves.



In our research, we are discovering that barriers to innovation are like gravity—pervasive, predictable, and not all that strong. There are many ways to overcome them, but the simplest and easiest path is to help your innovators notice what they’re overlooking. Often it’s right in front of their eyes.

Tony McCaffrey was formerly the chief technology officer of Innovation Accelerator. **Jim Pearson** was formerly the CEO of Innovation Accelerator.

Adapted from an article in *Harvard Business Review*, December 2015 (product #R1512F).

CHAPTER 13

Need Fresh Ideas? Call a Virtual Meeting

by Bob Frisch and Cary Greene

Andrew, the CEO of a \$500 million manufacturer, needed to pivot. His company's largest and most reliable revenue source had suddenly dried up. He thought that major changes in focus were required. His team would need to move quickly to sell a different mix of products to different customers.

Andrew set up a videoconference with his 12 direct reports and asked for their help in identifying pools of opportunity. With the press of a button, the executives found themselves in one of three virtual breakout rooms. Each group was tasked with highlighting five pressing market needs the company might be in a position to meet. Andrew had assigned people carefully and appointed leaders for each room; for example, he put the three most opinionated, extroverted leaders with the most forceful facilitator and left more-reserved reports with one another. Each team had a virtual whiteboard for capturing ideas. Andrew moved from one virtual room to another, spending a few minutes listening and commenting before providing some brief coaching points and moving on. After 30 minutes, the attendees were transported back to the main session.

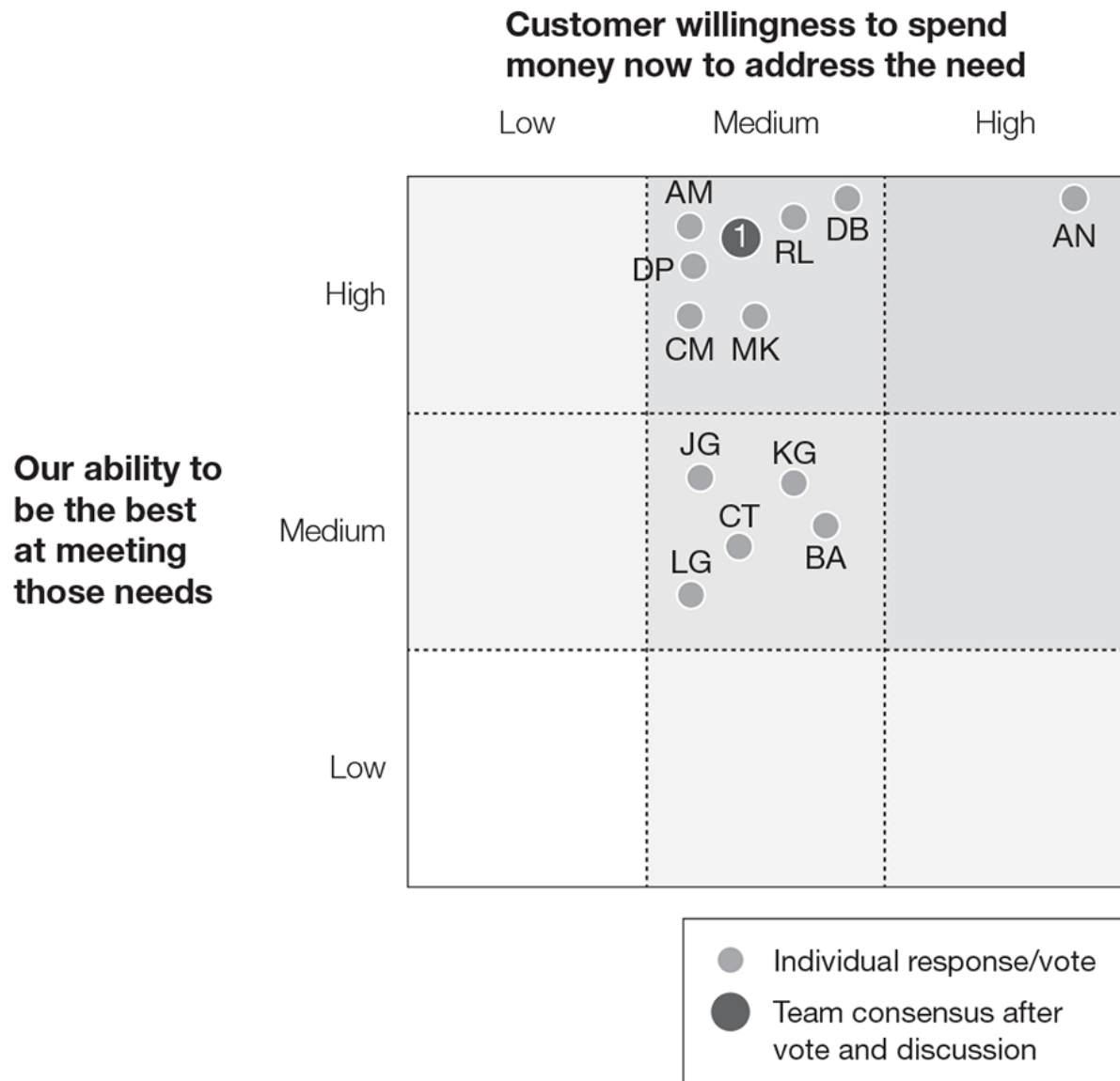
As the spokesperson from each breakout read out the group's top five potential opportunities, a master list of customer needs was captured live and appeared on every one's screen. A brief question-and-answer session ensured that the ideas were understood. Similar ones were consolidated. Some were refined.

Within an hour of starting the meeting, Andrew was staring at a good list: nine potentially promising areas for growth, four of which had never occurred to him. He shared his screen, displaying a three-by-three matrix of boxes, and asked team members to use the annotation function to initial the box in which they thought each new idea should fall. For the first idea, everyone landed in one of two boxes, as illustrated in [figure 13-1](#).

Andrew called on a few people to explain their placements, and after a quick conversation, the group agreed that the idea was “medium-high”—something they could do well but for which customer willingness to pay seemed mixed.

FIGURE 13-1

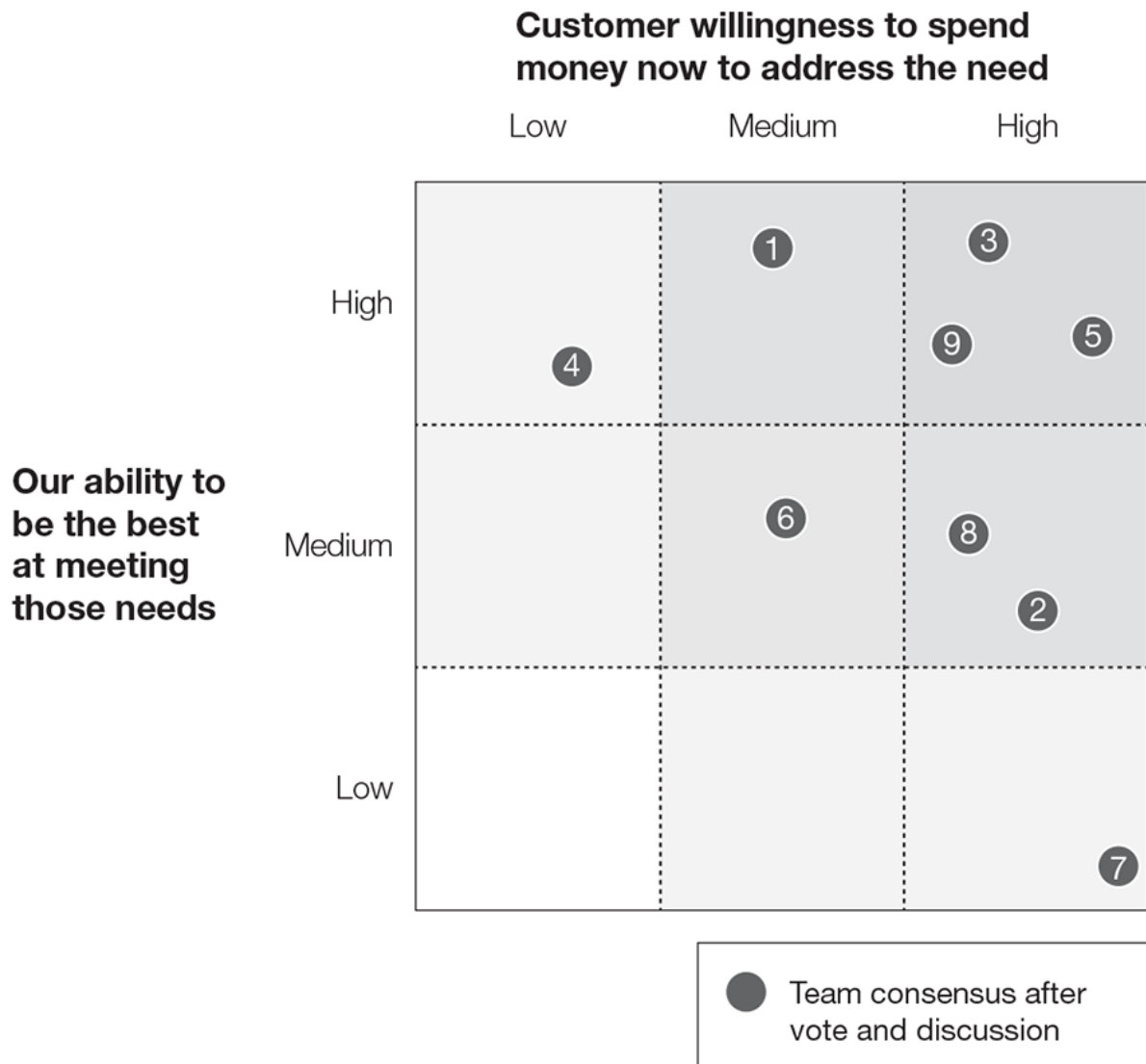
A team vote on where a new idea should fall along two criteria



Andrew then read the second idea and asked, “Relative to the first one, where should this idea be placed?” Team members again cast their votes. They repeated the exercise seven times, spending five to 10 minutes on each idea. If team members differed in their views, Andrew asked them to argue their preferences before determining a consensus placement. The result is shown in [figure 13-2](#).

FIGURE 13-2

A team vote on where nine total new ideas should fall along two criteria



After a 30-minute breakout, a 30-minute reporting period to refine and consolidate ideas, and 75 minutes of discussion, Andrew and his team had pinpointed the three best opportunities to pursue.

Could Andrew have held the same brainstorming session in person? Of course. But doing so would have been far less efficient. Just getting team members to and from breakout areas can take 10 to 15 minutes, and discussions can easily meander over an entire day without a clear conclusion. Three virtual meeting tools—the whiteboard, the breakout rooms, and the annotation function—combined to give Andrew more control, speed, structure, and clarity than he might have had sitting with his group around a conference table.

Bob Frisch is the founding partner of the Strategic Offsites Group. A regular contributor to *Harvard Business Review*, Bob wrote the bestselling *Who's in the Room? How Great Leaders Structure and Manage the Teams Around Them* and coauthored *Simple Sabotage*. He has earned over 10 million frequent-flier miles facilitating strategy meetings in 19 countries. **Cary Greene** is the managing partner of the Strategic Offsites Group. He is the coauthor of *Simple Sabotage* and a frequent contributor to *Harvard Business Review*, with articles featured across five collections, including *HBR Guide to Making Better Decisions*, *HBR Guide to Making Every Meeting Matter*, and *HBR Guide to Remote Work*.

Adapted from “3 Things Virtual Meetings Offer That In-Person Meetings Don’t” on hbr.org, July 23, 2020 (product #H05Q40).